

Определения (без фанатизма)

- **Черная дыра** – это область в пространстве-времени, где гравитационное притяжение настолько велико, что ни свет, ни материальные объекты не могут ее покинуть.
- **Горизонт событий** – это граница вокруг черной дыры, за пределами которой ничто не может вернуться назад. Это точка невозврата для любой материи и излучения.
- **Сингулярность** – это центральная точка внутри черной дыры, где плотность материи и кривизна пространства-времени становятся бесконечными. Физические законы, как мы их знаем, возможно перестают действовать в этой области.

Singularity

At the very centre of a black hole, matter has collapsed into a region of infinite density called a singularity.

All the matter and energy that fall into the black hole ends up here.

The prediction of infinite density by general relativity is thought to indicate the breakdown of the theory where quantum effects become important.

Event horizon

This is the radius around a singularity where matter and energy cannot escape the black hole's gravity: the point of no return.

This is the "black" part of the black hole.

Photon sphere

Although the black hole itself is dark, photons are emitted from nearby hot plasma in jets or an accretion disc (see below). In the absence of gravity, these photons would travel in straight lines, but just outside the event horizon of a black hole, gravity is strong enough to bend their paths so that we see a bright ring surrounding a roughly circular dark "shadow".

Relativistic jets

When a black hole feeds on stars, gas or dust, the meal produces jets of particles and radiation blasting out from the black hole's poles at near light speed.

They can extend for thousands of light-years into space.

Innermost stable orbit

The inner edge of an accretion disc is the last place that material can orbit safely without the risk of falling past the point of no return.

Accretion disc

A disc of superheated gas and dust whirls around a black hole at immense speeds, producing electromagnetic radiation (X-rays, optical, infrared and radio) that reveal the black hole's location. Some of this material is doomed to cross the event horizon, while other parts may be forced out to create jets.

Аккреционный
диск

Горизонт
событий

Струя газа

Сингулярность

Фотонная
сфера

Самая близкая
стабильная орбита